

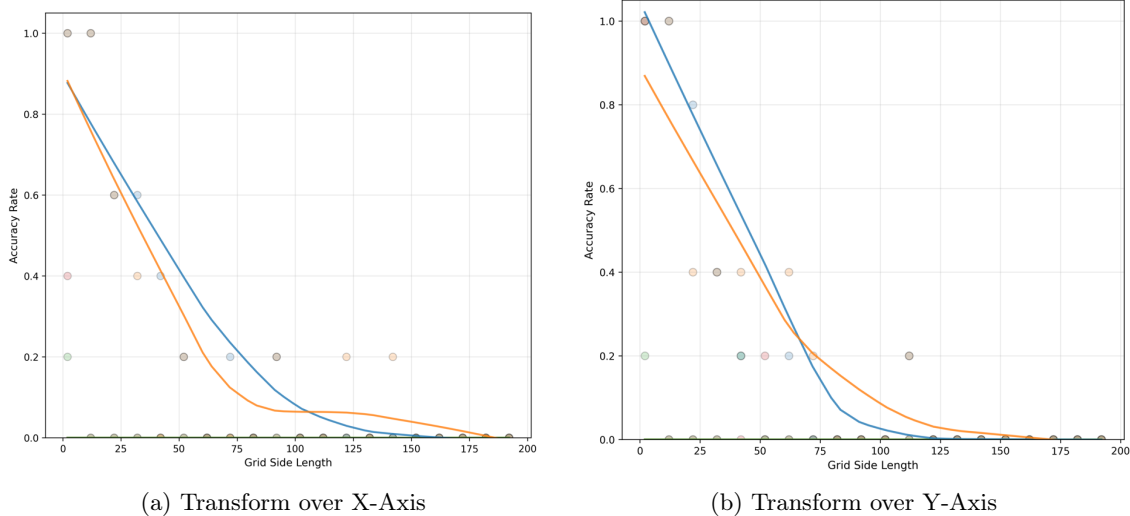


Figure 8: Accuracy vs Grid Size (Transformation)

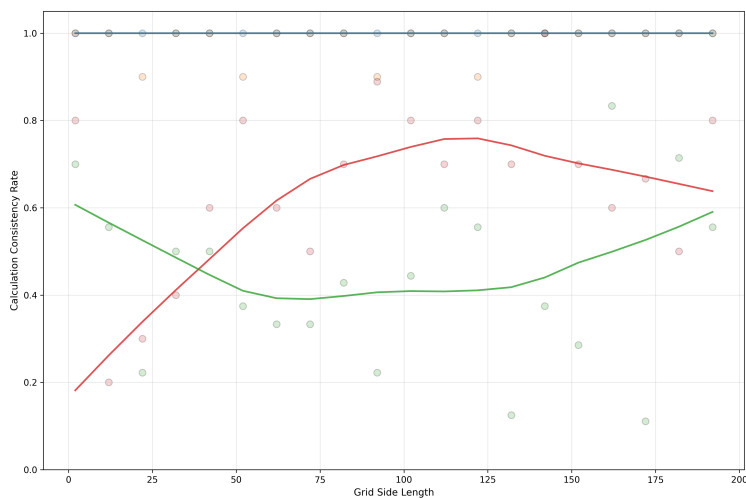


Figure 9: Consistency vs Grid Size (Transformation)

4.3 Distance

Models	Smallest 5 Grids	Largest 5 Grids
GPT-4o	44%	0%
GPT-4.1	76%	8%
Claude No Thinking	96%	40%
Claude Medium Thinking	100%	16%

Table 3: Model Accuracy on Small and Large Grids (Distance)

Given that the failures of the previous two tests were largely due to miscounting, we devised this test to evaluate a model’s ability to reason relatively; that is, in relation to other objects within a geometric space. Models were tested on grids ranging from 10×10 to 200×200 , with a step of 5. Like the previous tests, models

were more accurate on smaller boards and their performance degraded as grid size increased, as shown in Figure 10.

Interestingly, models did not exhibit much internal consistency, especially compared to the previous tests. They frequently made mathematical errors, accounting for around half of the errors for each model, as seen in Figure 11.

The math error rate decreased as grid size increased for the OpenAI models and remained roughly constant for the Anthropic models. The positional error rate increased with grid size, similar to other tests. The results are shown in Figure 12.

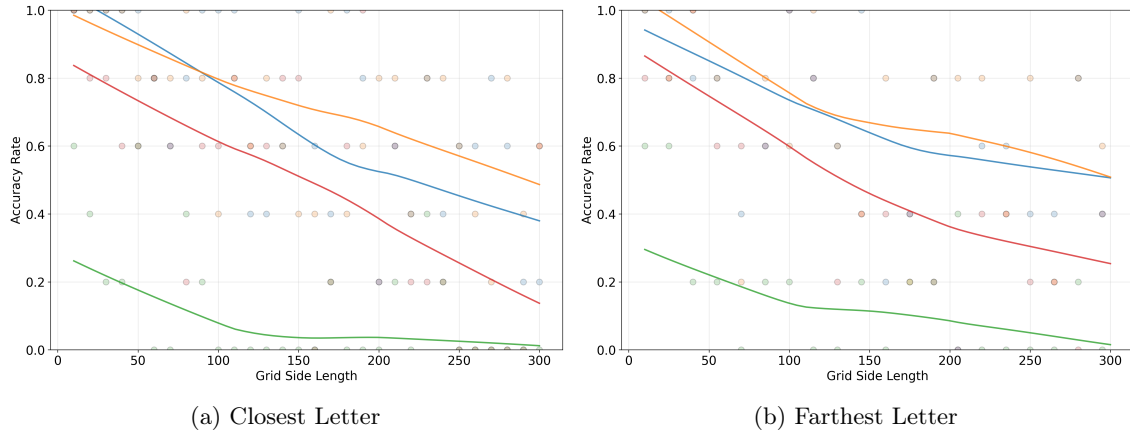


Figure 10: Accuracy vs Grid Size (Distance)

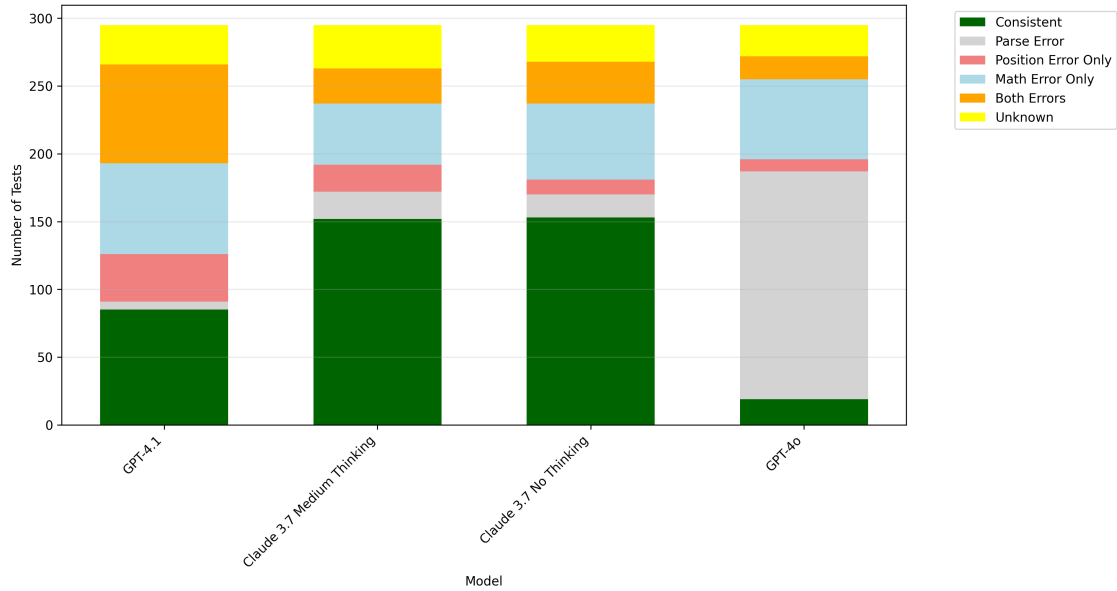


Figure 11: Error Types (Distance)

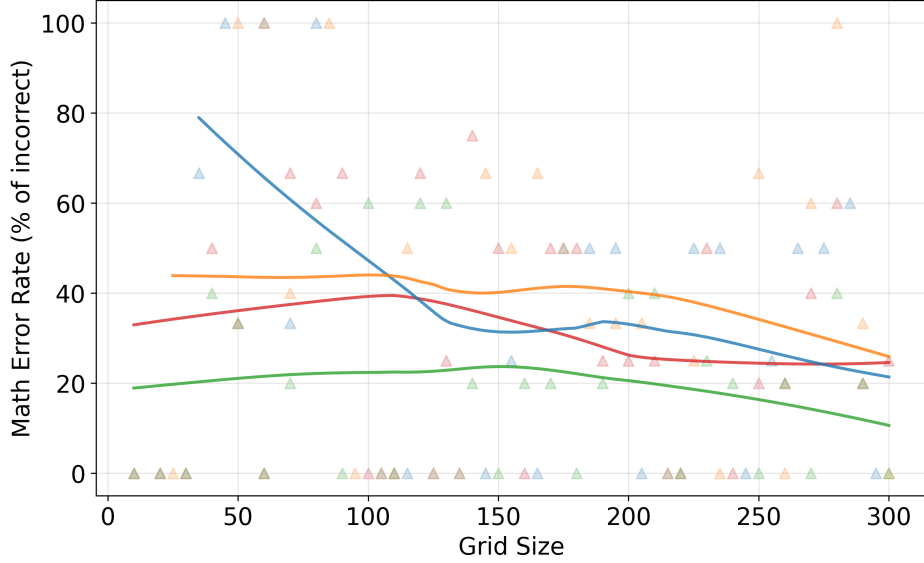


Figure 12: Math Error vs Grid Size (Distance)

4.4 Word Search

Models	Smallest 5 Grids	Largest 5 Grids
GPT-4o; 3 Letters	15%	0%
GPT-4.1; 3 Letters	72.5%	0%
Claude No Thinking; 3 Letters	95%	2%
Claude Medium Thinking; 3 Letters	95%	0%
GPT-4o; 6 Letters	0%	0%
GPT-4.1; 6 Letters	20%	0%
Claude No Thinking; 6 Letters	30%	0%
Claude Medium Thinking; 6 Letters	35%	0%
GPT-4o; 9 Letters	5.5%	0%
GPT-4.1; 9 Letters	47.2%	0%
Claude No Thinking; 9 Letters	80.5%	0%
Claude Medium Thinking; 9 Letters	72.2%	0%

Table 4: Model Accuracy on Small and Large Grids (Search)

This test was non-numerical in nature. Instead of calculations, we tested the models on language – a subject in which they traditionally excel. Grid sizes were tested from 5×5 to 100×100 with a step of 5, using words of length 3, 6, and 9. Words that did not fit in the board, ie. a nine-letter word for a 5×5 grid, were omitted from testing. All eight spelling directions (horizontal right-to-left, vertical up-to-down, etc) were tested for each word and grid size. Credit was given for percentage of coordinates correct, ranging from 0 to 1. This test was more difficult for the models from the start. The Anthropic models, which averaged close to 100% accuracy on smaller boards in the other tests, averaged only 50% for the six-letter words. The results for all three word lengths are seen in Figure 13.

Despite the low accuracy, however, the models still often claimed to have found the word. The Anthropic models had an almost 100% record of claimed detection, and the OpenAI models were somewhat lower. This was likely due to hallucination, the effects of which can be seen in Figure 14.